

An Equivariant, Pauli-Encoded Ansatz for the $\mathbb{Z}_2$ Lattice Gauge Theory

Compatible with Barthe’s Fourier Coefficient Extraction Algorithm

Internal research report

August 7, 2026

Abstract

We construct a parameterised quantum circuit for a $\mathbb{Z}_2$ lattice gauge theory (LGT) with matter that is simultaneously (i) *exactly equivariant* under an internal symmetry group G , so that the isotypic/Schur block-decoupling machinery of geometric quantum machine learning applies rigorously, and (ii) *exactly Pauli-encoded* in the sense of Barthe’s Definition 5.1, so that the Fourier Coefficient Extraction (FCE) algorithm applies without modification. The essential design decision is to build equivariance from *internal* symmetries only (global charge $U(1)_Q$ and the local Gauss law), and to discard the *spatial* symmetries (parity, charge conjugation, translation) and the gauge-inversion symmetry $\mathcal{Z}$, each of which we show is fatal for a different reason. We then prove a corrected sparsity theorem: symmetry does not shrink the ambient frequency lattice, it pins the frequency vector to a *sublattice*, via two mechanisms of very different strength — *additive* conservation laws, which reduce the *rank*, and *multiplicative* (parity) relations, which reduce only the *index*. In the $\mathbb{Z}_2$ model, Gauss’s law is multiplicative; we obtain a $2^{-(d+|V|)}$ reduction in the number of Fourier coefficients to be measured, but no change to the $d = O(\log n)$ threshold that governs efficiency. The analysis identifies precisely what would change the threshold: a gauge group whose Gauss law is *additive*, i.e. $U(1)$, for which the rank collapses from $|E|$ to the first Betti number $b_1(\Gamma)$ of the lattice graph. All derivations are given line by line and all notation is defined at first use.

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1 Purpose and executive summary

1.1 What problem we are solving

Barthe’s FCE algorithm gives an efficient quantum learner for a concept class built from Pauli-encoded parameterised quantum circuits (PQCs). Its efficiency hinges on one number: the size

$$m = |\Omega|$$

of the *Fourier frequency support* Ω of the learned function. Barthe’s generic bound is $m \leq (4L + 1)^d$, where d is the number of encoded parameters and L the number of times each is re-uploaded. This is polynomial only when $d = O(\log n)$, n being the number of qubits. Everything we do is aimed at the question:

Can the symmetries of a physical model be used to prove that Ω is much smaller than $(4L + 1)^d$, thereby weakening the $d = O(\log n)$ restriction?

1.2 The three findings

- F1. A viable equivariant ansatz exists, and costs nothing.** Taking $G = U(1)_Q \times \mathbb{Z}_{2,\text{Gauss}}^{|V|}$ — both *internal* symmetries — yields an ansatz that is exactly in the commutant $\rho(G)'$ and exactly Pauli-encoded. Remarkably, the equivariant generator set is the *physically correct* Trotterisation of the Hamiltonian; the non-equivariant version that a naive reading produces is the unphysical one (Section 5).
- F2. Three tempting symmetries must be discarded, each for a distinct reason** (Section 4). The spatial symmetries (parity P , charge conjugation C , translation T) act on the *classical input* x and would collapse the input space, destroying the learning problem. The gauge-inversion symmetry $\mathcal{Z} = \prod_e Z_e$ *annihilates* the electric term H_g under twirling, leaving a free-fermion model.
- F3. The correct sparsity statement is a sublattice theorem, not a dimension-counting theorem** (Section 6). Symmetry cannot shrink the ambient lattice $\{-2L, \dots, 2L\}^d$; it confines Ω to a sublattice Λ . Two mechanisms:

Relation type	Effect on Λ	Gain in $ \Omega $
Additive: $\sum_j c_j P_j = \kappa \mathbb{I}$	reduces rank by 1	factor $\Theta(L)$, and $d \rightarrow d - 1$
Multiplicative: $\prod_j P_j = \pm \mathbb{I}$	reduces index by 2	factor 2

In the $\mathbb{Z}_2$ model, Gauss’s law is *multiplicative* (it is a product of Paulis). We therefore obtain a large constant-factor gain ($2^{-(d+|V|)}$) but no change to the exponential scaling in d . Changing the gauge group to $U(1)$ makes Gauss’s law *additive*, and the rank then collapses to $b_1(\Gamma)$.

1.3 Bottom line for the PI

The $\mathbb{Z}_2$ model should be kept as the *validation platform* — it has finite-dimensional link Hilbert spaces, exact (Trotter-error-free) commuting blocks, and everything is checkable by $\mathbb{F}_2$ linear algebra in polynomial time. It should *not* be the model in which a learning separation is claimed. The separation target is the $U(1)$ theory, where the same theorem produces a genuine rank collapse.

2 The physical model, from scratch

This section assumes no background in high-energy physics or lattice gauge theory.

2.1 What a lattice gauge theory is, in one page

A *gauge theory* is a theory with a *local* symmetry: one may perform an independent symmetry transformation at every point of space, and the physics is unchanged. Electromagnetism is the

archetype: the electron field may be re-phased independently at every point, $\psi(r) \rightarrow e^{i\theta(r)}\psi(r)$, provided the electromagnetic potential is shifted correspondingly. The gauge field (the photon) exists precisely to compensate this local re-phasing.

Discretising space onto a graph $\Gamma = (V, E)$ gives a *lattice* gauge theory:

- **Matter** (electrons/quarks) lives on the **vertices** $v \in V$, also called *sites*.
- **The gauge field** (photons/gluons) lives on the **edges** $e \in E$, also called *links*. Intuitively, a link variable is the object that “carries” a matter particle from one site to the neighbouring one while keeping the local symmetry intact — it is a *parallel transporter*.
- **Not all states are physical.** Local symmetry implies a local constraint, **Gauss’s law**, which every physical state must obey. In electromagnetism this is $\nabla \cdot \vec{E} = \rho$: electric field lines can only begin and end on charges.

The *gauge group* says what kind of transformation is allowed at each site. For electromagnetism it is $U(1)$ (a continuous phase). The simplest possible choice is $\mathbb{Z}_2 = \{+1, -1\}$: the only allowed local transformation is a sign flip. This is our model. Its virtue is that a $\mathbb{Z}_2$ link variable has exactly two states, so **one link = one qubit**, with no truncation error. (A $U(1)$ link variable has infinitely many states and must be truncated — this is why we validate on $\mathbb{Z}_2$ first.)

2.2 Qubit encoding and notation

We work on a general connected graph $\Gamma = (V, E)$; the periodic chain of the reference paper is the special case $V = \{0, \dots, N_s - 1\}$, $E = \{(n, n+1)\}$, $|E| = |V| = N_s$. The total qubit count is

$$n = |V| + |E|.$$

Site qubits. One qubit per $v \in V$. In the Jordan–Wigner mapping used for staggered (Kogut–Susskind) fermions, the Pauli operator Z_v measures the local occupation: $Z_v = +1$ means the site is occupied, $Z_v = -1$ means empty. The ladder operators

$$\sigma_v^+ = \frac{1}{2}(X_v + iY_v), \quad \sigma_v^- = \frac{1}{2}(X_v - iY_v)$$

create and destroy a particle at v .

Link qubits. One qubit per $e \in E$. Two Pauli operators play distinct physical roles:

- X_e is the **electric field** on link e , with eigenvalues ± 1 (a $\mathbb{Z}_2$ electric flux is either present or absent). It is diagonal in the “electric basis”.
- Z_e is the **parallel transporter** (the $\mathbb{Z}_2$ analogue of the vector potential e^{iA}). Since $Z_e X_e = -X_e Z_e$, acting with Z_e *flips the electric flux* on link e , i.e. it creates or destroys one unit of electric flux there.

Symbol table. Used throughout; each is re-derived where it first appears.

Symbol	Meaning
$\Gamma = (V, E)$	lattice graph: vertices (sites) V , edges (links) E
$\delta(v)$	the set of edges incident to vertex v
Z_v	occupation of site v (+1 occupied, -1 empty)
X_e, Z_e	electric field on link e ; parallel transporter on link e
A_e, B_e	the two Pauli strings $X_v Z_e X_{v'}$ and $Y_v Z_e Y_{v'}$ for $e = (v, v')$
h_e	the gauge-covariant hopping operator $\frac{1}{2}(A_e + B_e)$; see §4.4
G_v	Gauss-law operator at v
Q	total charge $\sum_v Z_v$
Z	gauge inversion $\prod_e Z_e$
W_C	Wilson loop $\prod_{e \in C} Z_e$ on a cycle C
x	classical input of the learning problem (the graph / fixed-gate pattern)
$\alpha \in [0, 1]^d$	unknown parameters to be learned; the Fourier variable
d	number of encoded parameters
L	number of re-uploads (Trotter layers)
$l \in \mathbb{Z}^d$	a Fourier frequency vector; components l_j
$l_v^{(m)}, l_e^{(g)}, l_e^{(h)}$	components of l for the mass, electric and hopping parameters
Ω	the frequency support $\{l : b_l \neq 0\}$
$\mathbb{F}_2$	the two-element field $\{0, 1\}$ with arithmetic mod 2
$b_1(\Gamma)$	first Betti number $= E - V + 1$: number of independent loops

2.3 The Hamiltonian, term by term

$$H = J H_{\text{hop}} + m H_m + f H_g, \quad (1)$$

with $J, m, f \in \mathbb{R}$ the coupling constants (these are the physical quantities one would want to learn).

(1) Hopping term — kinetic energy and minimal coupling.

$$H_{\text{hop}} = \frac{1}{2} \sum_{e=(v,v') \in E} \left(\underbrace{X_v Z_e X_{v'}}_{=: A_e} + \underbrace{Y_v Z_e Y_{v'}}_{=: B_e} \right). \quad (2)$$

Physics. In fermionic language this is $\sum_e (\psi_v^\dagger U_e \psi_{v'} + \text{h.c.})$: a particle is destroyed at v' and created at v , and the link operator U_e (here Z_e) is inserted in between. That insertion is *minimal coupling*: the particle cannot move without dragging a unit of electric flux with it. This is what makes the theory a *gauge* theory rather than a free hopping model. The coefficient J plays the role of the propagation speed.

(2) Mass term — rest energy of the matter.

$$H_m = \sum_{v \in V} (-1)^v Z_v. \quad (3)$$

Physics. In the staggered-fermion formulation, particles live on even sites and *antiparticles* are holes on odd sites. The alternating sign $(-1)^v$ converts “occupation” into “rest-mass energy” uniformly for both species. Large m suppresses particle–antiparticle pair creation.

(3) Electric term — energy stored in the gauge field.

$$H_g = \sum_{e \in E} X_e. \quad (4)$$

Physics. This is the lattice analogue of $\int \vec{E}^2$. In $\mathbb{Z}_2$ the electric field squares to the identity, so the energy is linear in X_e . Large f makes electric flux expensive, so flux lines shorten: charges are pulled together. This is *confinement*. **H_g is the term that makes the model non-trivial:** it is the only term that does not commute with the link operators Z_e , and hence the only source of genuine gauge-field dynamics. Remember this — it is the crux of Section 4.2.

2.4 Gauss’s law

$$\boxed{G_v = Z_v \prod_{e \ni v} X_e} \quad (\text{on a chain: } G_v = X_{v-1,v} Z_v X_{v,v+1}). \quad (5)$$

Reading the formula. $\prod_{e \ni v} X_e$ is the product of the electric fields on all links touching v — the $\mathbb{Z}_2$ version of the divergence $\nabla \cdot \vec{E}$ (in $\mathbb{Z}_2$, “adding” means multiplying signs). Multiplying by Z_v , the local charge, gives exactly $\nabla \cdot \vec{E} - \rho$. A state is **physical** if

$$G_v |\psi\rangle = g_v |\psi\rangle \quad \forall v \in V, \quad g_v \in \{+1, -1\}.$$

The choice $g_v \equiv +1$ (or $g_v = (-1)^v$ with staggering conventions) is the charge-free sector; other sign patterns describe backgrounds of static external charges.

Three facts we will use constantly:

- (G1) The G_v are **Pauli operators** and **mutually commute**. (Check: G_v and $G_{v'}$ for adjacent v, v' overlap only on the shared link e , where both carry X_e ; identical Paulis commute. Sites are never shared.) Hence $\mathcal{S} := \langle G_v : v \in V \rangle$ is an abelian Pauli group — a **stabiliser group** — and the physical sector is a stabiliser code space. This is why every question about the sector reduces to $\mathbb{F}_2$ linear algebra in polynomial time.
- (G2) $[G_v, H_{\text{hop}}] = [G_v, H_m] = [G_v, H_g] = 0$ termwise, so gauge invariance is respected by each individual Pauli string. *No twirling is needed to enforce Gauss’s law.*
- (G3) G_v acts trivially on the lattice labels: it does not permute sites or links.

2.5 The global symmetries, and their status

Symmetry	Definition	Internal?	Kills a term?	Verdict
Gauss $\mathcal{S}$	$\langle G_v \rangle \cong \mathbb{Z}_2^{ V }$	yes	no	use
Charge $U(1)_Q$	$Q = \sum_v Z_v$	yes	no	use
Gauge inversion	$\mathcal{Z} = \prod_e Z_e$	yes	yes (H_g)	discard (§4.2)
Parity P	$v \mapsto V - v$	no	—	discard (§4.3)
Charge conj. C	$v \mapsto v + 1$ with signs	no	—	discard (§4.3)
Translation T	$v \mapsto v + 1$	no	—	discard (§4.3)

Here *internal* means: the symmetry acts on the Hilbert space but leaves the labels v, e fixed. *This distinction is the single most important design criterion in this report*, for reasons explained in §4.3.

We therefore adopt

$$G = U(1)_Q \times \mathbb{Z}_{2, \text{Gauss}}^{|V|} \quad (6)$$

which is **abelian**. Three consequences matter:

- (A1) All irreducible representations of an abelian group are one-dimensional, $d_\lambda = 1$. Hence in the isotypic decomposition $\mathcal{H} = \bigoplus_\lambda (\mathbb{C}^{m_\lambda} \otimes V_\lambda)$ the carrier spaces V_λ are trivial and $\mathcal{H}_\lambda = \mathbb{C}^{m_\lambda}$: *the isotypic components are just the joint eigenspaces of $(Q, \{G_v\})$* , i.e. the physical sectors labelled by $\lambda = (q, \{g_v\})$.
- (A2) The commutant is $\rho(G)' = \bigoplus_\lambda \mathcal{B}(\mathcal{H}_\lambda)$. Therefore *any* operator supported inside a single sector automatically lies in the commutant. This **repairs** the erroneous hypothesis “ $\rho_0 \in \rho(G)'$ ” flagged in the earlier review: for abelian G it holds automatically once $|\psi_0\rangle$ is a joint eigenvector.
- (A3) The spurious factor d_λ appearing in the earlier draft of the Fourier theorem disappears, since $d_\lambda = 1$.

3 What Barthe’s framework requires

3.1 Pauli encoding (Definition 5.1)

A circuit is *Pauli-encoded* if every parameterised gate has the form

$$V_{s,j}(\alpha) = e^{i\pi P_{s,j} \alpha_j}, \quad P_{s,j} \text{ a single Pauli string}, \quad (7)$$

with d parameters α_j , each re-uploaded L times ($s = 1, \dots, L$). Non-parameterised (“fixed”) gates are arbitrary. The crucial property is that a Pauli string P has spectrum exactly $\{+1, -1\}$.

3.2 Where the frequencies come from — derivation

This derivation is short, and everything later depends on it, so we do it explicitly.

Step 1 (spectral resolution of one gate). Write $P_j = (+1)\Pi_j^+ + (-1)\Pi_j^-$, where $\Pi_j^\pm$ are the orthogonal projectors onto the ± 1 eigenspaces. Then

$$e^{i\pi P_j \alpha_j} = \sum_{\epsilon=\pm 1} e^{i\pi \alpha_j \epsilon} \Pi_j^\epsilon. \quad (8)$$

So each encoding gate is a sum of two branches, each carrying a phase $e^{\pm i\pi \alpha_j}$.

Step 2 (expand the whole circuit). The circuit contains dL encoding gates, indexed by (s, j) . Expanding all of them,

$$U(x, \alpha)|0\rangle = \sum_{\vec{\epsilon}} \exp\left(i\pi \sum_{s=1}^L \sum_{j=1}^d \alpha_j \epsilon_{s,j}\right) K_{\vec{\epsilon}}(x)|0\rangle, \quad (9)$$

where $\vec{\epsilon} = (\epsilon_{s,j}) \in \{\pm 1\}^{L \times d}$ is a *branch assignment* (which eigenvalue was selected at each gate), and $K_{\vec{\epsilon}}(x)$ is the corresponding product of projectors and fixed gates.

Step 3 (define the accumulated frequency). The phase depends on $\vec{\epsilon}$ only through

$$\Lambda_j(\vec{\epsilon}) := \sum_{s=1}^L \epsilon_{s,j} \in \mathbb{Z}, \quad j = 1, \dots, d. \quad (10)$$

Interpretation. Λ_j is a **counter**. Every time the j -th encoding gate acts, the counter moves by $+1$ or -1 according to which eigenvalue branch the state is in. This is exactly what Barthe's algorithm $\mathcal{A}$ implements physically: it adds a *frequency register* and replaces each encoding gate by a controlled *increment/decrement* on that register. The phrase “*frequency shift*” means precisely: the ± 1 step taken by this counter at one gate.

Since Λ_j is a sum of L terms each equal to ± 1 :

$$\Lambda_j \in \{-L, -L+2, \dots, L-2, L\}, \quad \text{in particular } \Lambda_j \equiv L \pmod{2}. \quad (11)$$

Step 4 (expectation values). The learned function is $f_x(\alpha) = \langle 0|U^\dagger(x, \alpha) \mathcal{O} U(x, \alpha)|0\rangle$. The bra carries its own branch assignment $\vec{\epsilon}'$ with counter Λ' , and contributes the conjugate phase. The net Fourier frequency is the *difference of the two counters*:

$$l = \Lambda - \Lambda' \in \mathbb{Z}^d, \quad f_x(\alpha) = \sum_l b_l(x) e^{i\pi l \cdot \alpha}. \quad (12)$$

Step 5 (range and parity — a free improvement). From (11), $l_j = \Lambda_j - \Lambda'_j$ satisfies $|l_j| \leq 2L$, and moreover

$$l_j \equiv L - L \equiv 0 \pmod{2}: \quad \text{every frequency component is even.} \quad (13)$$

Barthe indexes b by $\{-2L, \dots, 2L\}^d$, of cardinality $(4L+1)^d$. Only the $(2L+1)^d$ all-even points can be non-zero. *Before any symmetry argument at all, a factor $\approx 2^d$ of the coefficients are identically zero.* Relabelling $l \rightarrow l/2$ saves one qubit per frequency sub-register and shrinks the LASSO design matrix by 2^d .

3.3 The roles of x and α — do not confuse them

In Barthe's Definition 5.2:

- $\alpha \in [0, 1]^d$ are the **unknown** parameters. They are the Pauli-encoded ones. The frequencies l are conjugate to α .
- x is the **classical input**: the bitstring specifying the *fixed* gates (for us: the graph Γ , or a disorder pattern). It enters only through the coefficients $b_l(x)$.
- The concept is $c_\alpha : x \mapsto \sum_l b_l(x) e^{i\pi l \cdot \alpha}$. The learner sees pairs (x_t, y_t) for one fixed unknown α^* , uses FCE to obtain the feature vector $b(x_t)$, and runs LASSO in the weights $w_l = e^{i\pi l \cdot \alpha^*}$.
- The classical hardness (Barthe Thm. 5.3) comes entirely from the x -dependence: some x encodes a BQP-hard instance.

Consequence, and the reason for our design principle: a symmetry that acts on the lattice *labels* acts on x . Demanding equivariance under such a symmetry restricts the admissible x , and therefore attacks the only source of hardness in the construction.

4 The twirl: definition, and which symmetries survive it

4.1 Definition and the projector property

Definition 4.1 (Twirl). For a compact group G with unitary representation ρ , the *twirl* of an operator A is

$$\mathcal{T}_G(A) = \int_G d\mu(g) \rho(g) A \rho(g)^\dagger \quad \left(= \frac{1}{|G|} \sum_{g \in G} \rho(g) A \rho(g)^\dagger \text{ for finite } G \right), \quad (14)$$

with μ the normalised Haar measure.

Lemma 4.2 (Twirl = orthogonal projector onto the commutant). $\mathcal{T}_G$ is the orthogonal projector (in the Hilbert–Schmidt inner product) onto $\rho(G)' = \{A : [A, \rho(g)] = 0 \ \forall g\}$. In particular: (a) $\mathcal{T}_G(A) \in \rho(G)'$; (b) $\mathcal{T}_G(A) = A \iff A \in \rho(G)'$; (c) $\mathcal{T}_G^2 = \mathcal{T}_G$; (d) if $\rho(g_0) A \rho(g_0)^\dagger = -A$ for some g_0 , then $\mathcal{T}_G(A) = 0$.

Proof. (a) For any $h \in G$, substituting $g \rightarrow hg$ and using invariance of Haar measure, $\rho(h) \mathcal{T}_G(A) \rho(h)^\dagger = \int d\mu(g) \rho(hg) A \rho(hg)^\dagger = \mathcal{T}_G(A)$. (b) If A commutes with every $\rho(g)$ the integrand is constantly A . Conversely (a) gives the reverse inclusion. (c) follows from (a)+(b). (d) Group the g -integral into cosets of $\langle g_0 \rangle$; each coset contributes $\frac{1}{2}(\rho(g) A \rho(g)^\dagger + \rho(gg_0) A \rho(gg_0)^\dagger)$. Writing $\rho(gg_0) A \rho(gg_0)^\dagger = \rho(g)(\rho(g_0) A \rho(g_0)^\dagger) \rho(g)^\dagger = -\rho(g) A \rho(g)^\dagger$, each coset contributes zero. $\square$

Item (d) is the mechanism that destroys H_g , so we now make it explicit.

4.2 Why the $\mathcal{Z}$ -twirl annihilates the electric term

Let $\mathcal{Z} = \prod_{e' \in E} Z_{e'}$. It is Hermitian and unitary with $\mathcal{Z}^2 = \mathbb{I}$, so $\langle \mathcal{Z} \rangle = \{\mathbb{I}, \mathcal{Z}\} \cong \mathbb{Z}_2$ and the twirl is the two-term average $\mathcal{T}(A) = \frac{1}{2}(A + \mathcal{Z} A \mathcal{Z}^\dagger)$.

Step 1 — split off the relevant factor. Fix a link e . Write

$$\mathcal{Z} = Z_e \cdot \prod_{e' \neq e} Z_{e'}.$$

The second factor acts only on link qubits $e' \neq e$, which are disjoint from the qubit e on which X_e acts. Operators on disjoint qubits commute, so that factor cancels against its inverse:

$$\mathcal{Z} X_e \mathcal{Z}^\dagger = Z_e X_e Z_e^\dagger = Z_e X_e Z_e. \quad (15)$$

Step 2 — single-qubit Pauli algebra. Using $ZX = iY$ and $YZ = iX$:

$$ZXZ = (ZX)Z = (iY)Z = i(YZ) = i(iX) = -X.$$

Hence

$$\mathcal{Z} X_e \mathcal{Z}^\dagger = -X_e. \quad (16)$$

In words: the electric field operator is **odd** under global gauge inversion.

Step 3 — twirl.

$$\boxed{\mathcal{T}_{\langle \mathcal{Z} \rangle}(X_e) = \frac{1}{2}(X_e + (-X_e)) = 0 \implies \mathcal{T}_{\langle \mathcal{Z} \rangle}(H_g) = 0.} \quad (17)$$

Step 4 — why this is fatal, not cosmetic. With H_g deleted, the surviving generators are $\{A_e, B_e, Z_v\}$. Every link operator Z_e now commutes with all of them: $[Z_e, A_e] = 0$ (both

carry Z on link e ; site qubits are disjoint), $[Z_e, B_e] = 0$ likewise, and $[Z_e, Z_v] = 0$ (disjoint qubits). Therefore *every* Z_e is a constant of motion: the link variables freeze to fixed classical signs $z_e = \pm 1$. Substituting $Z_e \rightarrow z_e$ in (2) gives

$$H \longrightarrow \frac{J}{2} \sum_{e=(v,v')} z_e (X_v X_{v'} + Y_v Y_{v'}) + m \sum_v (-1)^v Z_v,$$

which is an XY spin chain with modulated bond signs plus a field. Under Jordan–Wigner this is **quadratic in fermions** — a free-fermion / matchgate model, classically simulable in polynomial time. Barthe’s hardness theorem requires a BQP-hard concept in the class; a matchgate family cannot supply one. Enforcing $\mathcal{Z}$ therefore destroys the learning separation before any learning question is posed.

Moral: H_g is the only term generating gauge-field dynamics; a symmetry under which it is odd cannot be imposed.

4.3 Why the spatial symmetries P , C , T must also go

Two independent reasons.

(i) **The equivariance dilemma.** The Schur/isotypic decoupling theorem requires the *whole* circuit — including the fixed gates that encode x — to lie in $\rho(G)'$. If G contains P or T , then the fixed-gate layer must itself be reflection/translation invariant, so x ranges only over symmetric configurations. On a chain this leaves $O(1)$ admissible inputs, and the concept class collapses to a constant-size family, brute-force learnable classically. If instead one imposes equivariance only on the encoding layer, the circuit is not in the commutant and the Schur theorem does not apply — so the claimed sparsity has no proof. Either horn is fatal.

(ii) **C is not a $\mathbb{Z}_2$.** With the standard definition ($X_v \mapsto X_{v+1}$, $Y_v, Z_v \mapsto -Y_{v+1}, -Z_{v+1}$), C generates a group of order $|V|$ (translation with signs), not of order 2. Its twirl is therefore a *full translation average*, which maps $Z_v \mapsto \frac{1}{|V|} H_m$ and $A_e \mapsto \frac{1}{|V|} H_{\text{hop}}^{XX}$. The number of independent encoded parameters collapses from $O(|V|)$ to $d = O(1)$: one recovers the homogeneous Hamiltonian $\{J, m, f\}$ and loses all input dependence.

4.4 The $U(1)_Q$ twirl — full derivation

This is the constructive part. Let $Q = \sum_v Z_v$ (site qubits only) and

$$R(\phi) = e^{i\phi Q/2} = \prod_v e^{i\phi Z_v/2}, \quad \phi \in [0, 2\pi).$$

The factor $1/2$ is a normalisation chosen so that ladder operators pick up $e^{\pm i\phi}$ exactly.

Step 1 — action on ladder operators. From $[Z_v, \sigma_v^\pm] = \pm 2\sigma_v^\pm$,

$$\frac{d}{d\phi} \left(e^{i\phi Z_v/2} \sigma_v^\pm e^{-i\phi Z_v/2} \right) = \frac{i}{2} e^{i\phi Z_v/2} [Z_v, \sigma_v^\pm] e^{-i\phi Z_v/2} = \pm i \left(e^{i\phi Z_v/2} \sigma_v^\pm e^{-i\phi Z_v/2} \right),$$

a first-order ODE with solution

$$R(\phi) \sigma_v^\pm R(\phi)^\dagger = e^{\pm i\phi} \sigma_v^\pm. \quad (18)$$

Meaning. σ^+ adds one unit of charge and is therefore rotated by one unit of phase; this is what “ $U(1)$ charge symmetry” means operationally.

Step 2 — rewrite A_e in ladder operators. For $e = (v, v')$, using $X = \sigma^+ + \sigma^-$,

$$A_e = X_v Z_e X_{v'} = (\sigma_v^+ + \sigma_v^-) Z_e (\sigma_{v'}^+ + \sigma_{v'}^-). \quad (19)$$

Note Z_e is a *link* operator: $R(\phi)$ acts only on site qubits, so Z_e is untouched.

Step 3 — conjugate. Applying (18) to each factor:

$$\begin{aligned} R(\phi) A_e R(\phi)^\dagger &= (e^{i\phi} \sigma_v^+ + e^{-i\phi} \sigma_v^-) Z_e (e^{i\phi} \sigma_{v'}^+ + e^{-i\phi} \sigma_{v'}^-) \\ &= e^{2i\phi} \sigma_v^+ Z_e \sigma_{v'}^+ + \sigma_v^+ Z_e \sigma_{v'}^- + \sigma_v^- Z_e \sigma_{v'}^+ + e^{-2i\phi} \sigma_v^- Z_e \sigma_{v'}^-. \end{aligned} \quad (20)$$

The four terms have definite *charge transfer*: $+2, 0, 0, -2$.

Step 4 — average over the group. Using $\frac{1}{2\pi} \int_0^{2\pi} e^{ik\phi} d\phi = \delta_{k,0}$ for $k \in \mathbb{Z}$, the $e^{\pm 2i\phi}$ terms integrate to zero:

$$\mathcal{T}_{U(1)}(A_e) = \sigma_v^+ Z_e \sigma_{v'}^- + \sigma_v^- Z_e \sigma_{v'}^+. \quad (21)$$

Step 5 — recognise the result as $\frac{1}{2}(A_e + B_e)$. Expand both products:

$$\begin{aligned} X_v X_{v'} &= (\sigma_v^+ + \sigma_v^-)(\sigma_{v'}^+ + \sigma_{v'}^-) = \sigma_v^+ \sigma_{v'}^+ + \sigma_v^+ \sigma_{v'}^- + \sigma_v^- \sigma_{v'}^+ + \sigma_v^- \sigma_{v'}^-, \\ Y_v Y_{v'} &= [-i(\sigma_v^+ - \sigma_v^-)] [-i(\sigma_{v'}^+ - \sigma_{v'}^-)] = -(\sigma_v^+ \sigma_{v'}^+ - \sigma_v^+ \sigma_{v'}^- - \sigma_v^- \sigma_{v'}^+ + \sigma_v^- \sigma_{v'}^-). \end{aligned}$$

Adding, the charge-changing terms cancel and the charge-conserving ones double:

$$X_v X_{v'} + Y_v Y_{v'} = 2(\sigma_v^+ \sigma_{v'}^- + \sigma_v^- \sigma_{v'}^+). \quad (22)$$

Since Z_e commutes with all site operators, inserting it and comparing with (21):

$$\boxed{\mathcal{T}_{U(1)}(A_e) = \frac{1}{2}(A_e + B_e) =: h_e.} \quad (23)$$

Step 6 — what is $\mathcal{T}_{U(1)}(B_e)$? The same operator h_e . Indeed $R(\phi) Y_v R(\phi)^\dagger = -i(e^{i\phi} \sigma_v^+ - e^{-i\phi} \sigma_v^-)$, so

$$R(\phi) B_e R(\phi)^\dagger = -\left(e^{2i\phi} \sigma_v^+ Z_e \sigma_{v'}^+ - \sigma_v^+ Z_e \sigma_{v'}^- - \sigma_v^- Z_e \sigma_{v'}^+ + e^{-2i\phi} \sigma_v^- Z_e \sigma_{v'}^-\right),$$

and averaging kills the $e^{\pm 2i\phi}$ terms and leaves $\sigma_v^+ Z_e \sigma_{v'}^- + \sigma_v^- Z_e \sigma_{v'}^+ = h_e$. This is a useful consistency check: A_e and B_e have the same charge-conserving part, and applying the twirl to h_e itself returns h_e (idempotence, Lemma 4.2(c)).

Step 7 — physical meaning of h_e . From (21),

$$h_e = \underbrace{\sigma_v^+}_{\text{create particle at } v} \underbrace{Z_e}_{\text{flip flux on } e} \underbrace{\sigma_{v'}^-}_{\text{destroy particle at } v'} + \text{h.c.}$$

This is exactly the **gauge-covariant hopping** (minimal coupling): one unit of charge moves along the link, and the parallel transporter simultaneously flips the electric flux on that link — which is precisely what Gauss's law demands, since moving a charge must move the endpoint of a flux line. The terms discarded by the twirl, $\sigma^+ \sigma^+$ and $\sigma^- \sigma^-$, create or destroy *pairs* of particles; they change Q by ± 2 and are exactly what breaks $U(1)$.

Step 8 — the other two generators are untouched. $[Q, Z_v] = 0$ since Z commutes with Z ; and $[Q, X_e] = 0$ since Q acts on site qubits while X_e acts on a link qubit (disjoint supports). Hence

$$\mathcal{T}_{U(1)}(Z_v) = Z_v, \quad \mathcal{T}_{U(1)}(X_e) = X_e. \quad (24)$$

Nothing is annihilated — contrast §4.2. And by (G2) the Gauss twirl acts as the identity on all three generators, so no further work is needed.

5 The equivariant, Pauli-encoded ansatz

5.1 Generator set

$$\Gamma_{\text{enc}} = \underbrace{\{Z_v\}_{v \in V}}_{\text{mass}} \cup \underbrace{\{X_e\}_{e \in E}}_{\text{electric}} \cup \underbrace{\{h_e = \frac{1}{2}(A_e + B_e)\}_{e \in E}}_{\text{hopping}} \subset \rho(G)'. \quad (25)$$

Proposition 5.1 (Key structural facts). (i) $[A_e, B_e] = 0$.

(ii) $\text{spec}(h_e) = \{-1, 0, 0, +1\}$ — *integer, as FCE requires.*

(iii) $e^{i\pi\alpha(A_e+B_e)} = e^{i\pi\alpha A_e} e^{i\pi\alpha B_e}$ exactly, *with no Trotter error.*

Proof. (i) Two Pauli strings commute iff they anticommute on an *even* number of qubits. A_e and B_e overlap on three qubits: site v (X vs. Y : anticommute), link e (Z vs. Z : commute), site v' (X vs. Y : anticommute). Two anticommutations, an even number, hence $[A_e, B_e] = 0$.

(ii) Since A_e, B_e commute they have a joint eigenbasis with eigenvalues $(\epsilon_A, \epsilon_B) \in \{\pm 1\}^2$, giving h_e eigenvalues $\frac{1}{2}(\epsilon_A + \epsilon_B) \in \{-1, 0, +1\}$. All four sign patterns actually occur provided $A_e B_e \neq \pm \mathbb{I}$. Compute

$$A_e B_e = (X_v Z_e X_{v'}) (Y_v Z_e Y_{v'}) = (X_v Y_v) (Z_e Z_e) (X_{v'} Y_{v'}) = (iZ_v) \mathbb{I} (iZ_{v'}) = -Z_v Z_{v'},$$

which is not $\pm \mathbb{I}$. Hence $\text{spec}(h_e) = \{-1, 0, 0, +1\}$, with the eigenvalue 0 doubly degenerate.

(iii) Immediate from (i): commuting exponents split exactly. $\square$

Remark 5.2 (Equivariance is free). The physical first-order Trotterisation of JH_{hop} assigns A_e and B_e the *same* angle $J\tau/r$, because they come from the same Hamiltonian term with the same coupling. Treating them as two *independent* encoded parameters — which is what a naive reading of “one Pauli string, one parameter” produces — is the unphysical choice, and it is exactly the choice that breaks $U(1)_Q$. **We are not paying a price for equivariance; we are simply not gratuitously breaking the model.**

5.2 Circuit form and Pauli-encoding compliance

One Trotter layer $s = 1, \dots, L$:

$$W_s(x, \alpha) = V_s^{\text{fix}}(x) \underbrace{\prod_{v \in V} e^{i\pi\alpha_v^{(m)} Z_v} \prod_{e \in E} e^{i\pi\alpha_e^{(g)} X_e}}_{\text{commuting block } F} \prod_{e \in E} \left[e^{i\pi\alpha_e^{(h)} A_e} e^{i\pi\alpha_e^{(h)} B_e} \right]. \quad (26)$$

Every parameterised gate is $e^{i\pi P \alpha_j}$ with P a *single* Pauli string, so Definition 7 is met verbatim. The only departure from the plain setting is **parameter tying**: $\alpha_e^{(h)}$ drives two gates. Barthe’s algorithm $\mathcal{A}$ handles this unchanged — it simply emits two increment/decrement operations onto the *same* frequency sub-register.

We write r_j for the number of gates driven by parameter j in one layer (the *multiplicity*): $r_j = 1$ for mass and electric parameters, $r_j = 2$ for hopping parameters. The parameter count is

$$d = |V| + |E| + |E| = |V| + 2|E|. \quad (27)$$

5.3 Register sizing — get this wrong and you will alias

Repeating the derivation of §3.2 with multiplicity r_j : the counter Λ_j now receives r_j steps of ± 1 per layer, hence $r_j L$ steps in total.

Coordinate	count	r_j	Λ_j range	l_j range	# admissible l_j
$\alpha_v^{(m)}$ (mass)	$ V $	1	$\{-L, \dots, L\}$	even, $\{-2L, \dots, 2L\}$	$2L + 1$
$\alpha_e^{(g)}$ (electric)	$ E $	1	$\{-L, \dots, L\}$	even, $\{-2L, \dots, 2L\}$	$2L + 1$
$\alpha_e^{(h)}$ (hopping)	$ E $	2	$\{-2L, \dots, 2L\}$	even, $\{-4L, \dots, 4L\}$	$4L + 1$

(The parity claim (13) still holds: Λ_j is a sum of $r_j L$ terms ± 1 , so $\Lambda_j \equiv r_j L \pmod{2}$; the difference of two such is even.)

Consequently:

- **State register** width per coordinate: $\lceil \log_2(2r_j L + 1) \rceil$ (Barthe’s $\lceil \log_2(2L + 1) \rceil$ is the case $r_j = 1$).
- **Frequency register** width per coordinate: $\lceil \log_2(4r_j L + 1) \rceil$.
- Total: $n' = n + \sum_{j=1}^d \lceil \log_2(4r_j L + 1) \rceil + 1$, a factor ≤ 2 over Barthe.
- Gate count: each parameterised gate contributes $O(n)$ basis-change gates plus one controlled increment/decrement, and there are $\sum_j r_j = |V| + 3|E|$ of them per layer instead of $d = |V| + 2|E|$.

Warning. If the hopping sub-registers are given Barthe’s default width $\lceil \log_2(2L + 1) \rceil$, the counter wraps around modulo the register size and the extracted coefficients are *aliased sums* — silently wrong, with no error flag. Halving the labels ($l \rightarrow l/2$, legitimate by (13)) recovers one qubit per coordinate.

5.4 Initial state

We need $|\psi_0\rangle$ to be a joint eigenvector of Q and all G_v , so that it sits inside a single isotypic component $\lambda_0 = (q, \{g_v\})$.

Site register. Take the staggered “bare vacuum”, alternating occupied/empty: $|0101 \dots\rangle_{\text{sites}}$. Then Z_v has definite eigenvalue $z_v = (-1)^{v+1}$ and

$$Q|\psi_0\rangle = \left(\sum_v z_v \right) |\psi_0\rangle = 0 \cdot |\psi_0\rangle,$$

for even $|V|$: the charge-neutral sector, $q = 0$.

Link register. We must solve $G_v|\psi_0\rangle = g_v|\psi_0\rangle$, i.e.

$$\prod_{e \ni v} x_e = g_v z_v \quad \forall v \in V, \tag{28}$$

where $x_e = \pm 1$ is the X_e eigenvalue. Writing $x_e = (-1)^{\xi_e}$ with $\xi \in \mathbb{F}_2^E$, this is a *linear system over $\mathbb{F}_2$* ,

$$B \xi = \beta, \quad B = \text{vertex-edge incidence matrix over } \mathbb{F}_2,$$

solvable by Gaussian elimination in $O(|V||E|^2)$ time. On a single cycle it is solvable iff $\prod_v (g_v z_v) = +1$, and then it has exactly two solutions differing by the global flip $\xi \rightarrow \xi + \mathbf{1}$ — which is precisely the gauge-inversion symmetry $\mathcal{Z}$. Taking the symmetric combination gives the reference paper’s state

$$|\psi_0\rangle = |0101\cdots\rangle_{\text{sites}} \otimes \frac{1}{\sqrt{2}} \left(|++\cdots+\rangle + |--\cdots-\rangle \right)_{\text{links}}, \quad (29)$$

which is additionally a $\mathcal{Z}$ eigenstate.

Why this repairs the earlier theorem. Because G is abelian, its irreps are one-dimensional and its isotypic components are the joint eigenspaces $\mathcal{H}_\lambda$. The commutant is $\rho(G)' = \bigoplus_\lambda \mathcal{B}(\mathcal{H}_\lambda)$, so *any* operator supported in a single $\mathcal{H}_{\lambda_0}$ — in particular $\rho_0 = |\psi_0\rangle\langle\psi_0|$ — lies in $\rho(G)'$ automatically. The hypothesis flagged as false in the earlier (non-abelian) formulation now holds by construction.

5.5 Observables, and the Wilson loop

The observable must also be equivariant. Admissible choices: Z_v ; the staggered mass density $\sum_v (-1)^v Z_v$; and — the physically meaningful one — the **Wilson loop**

$$W_C = \prod_{e \in C} Z_e, \quad (30)$$

where $C \subseteq E$ is a *cycle*: a set of edges meeting every vertex an even number of times (equivalently, C is a closed loop, or a disjoint union of closed loops). The Wilson loop is the standard order parameter for confinement: its expectation decays with the enclosed *area* in the confined phase and with the *perimeter* in the deconfined phase.

Lemma 5.3. $[W_C, G_v] = 0$ for all $v \in V$, and $[W_C, Q] = 0$.

Proof. (a) $[W_C, Q] = 0$. W_C is a product of Z_e on *link* qubits; $Q = \sum_v Z_v$ acts on *site* qubits. Their supports are disjoint, and operators on disjoint qubits commute. (This is the sense in which the statement is “trivial”.)

(b) $[W_C, G_v] = 0$. Both are Pauli strings, so we count anticommuting overlaps. $G_v = Z_v \prod_{e \ni v} X_e$ acts by Z on the site qubit v and by X on each link in $\delta(v)$. W_C acts by Z on each link in C and not at all on site qubits. Hence:

- *Site qubit v* : W_C acts as identity there. No contribution.
- *Link $e \in C \cap \delta(v)$* : W_C has Z_e , G_v has X_e ; Z and X anticommute. Contributes 1.
- *Link $e \in C \setminus \delta(v)$* : G_v acts as identity there. No contribution.
- *Link $e \in \delta(v) \setminus C$* : W_C acts as identity there. No contribution.

So the number of anticommuting overlaps is exactly $|C \cap \delta(v)|$, the number of edges of the loop C incident to v . By the definition of a cycle, this number is **even** for every v (for a simple closed loop it is 2 if the loop passes through v and 0 otherwise). An even number of anticommutations means the two Pauli strings commute. $\square$

Remark 5.4. Lemma 5.3 is the first appearance of the graph-topological structure that will dominate §7.5: “even at every vertex” is exactly the definition of an $\mathbb{F}_2$ cycle.

6 The sparsity theorem

6.1 What we are claiming, in words

The earlier draft claimed that symmetry *shrinks* the frequency spectrum, bounding $|\Omega| \leq \sum_{\lambda} m_{\lambda}^{2L}$. That is not the right statement: since the encoding generators still have ± 1 spectra after restriction to a sector, the ambient lattice $\{-2L, \dots, 2L\}^d$ is unchanged. What symmetry actually does is impose **linear constraints** on the frequency vector l , confining Ω to a *sublattice* $\Lambda \subseteq \mathbb{Z}^d$.

There are exactly two kinds of constraint, of very different strength:

	Additive relation	Multiplicative relation
Form on the sector	$\sum_j c_j P_j = \kappa \mathbb{I}$	$\prod_{j \in S} P_j = \pm \mathbb{I}$
Constraint on l	$c \cdot l = 0$ (an integer hyperplane)	$\sum_{j \in S} \frac{l_j}{2} \equiv 0 \pmod{2}$ (an $\mathbb{F}_2$ parity check)
Effect	kills one dimension : $\text{rank } \Lambda \rightarrow \text{rank } \Lambda - 1$	kills one bit : index of Λ doubles
Gain in $ \Omega $	factor $\Theta(L)$, and <i>effectively</i> $d \rightarrow d - 1$	factor 2

Why the distinction is decisive. $|\Omega| \approx (\text{range})^{\text{rank } \Lambda}$. Removing a dimension changes the exponent; removing a bit changes only a prefactor. To move Barthe's $d = O(\log n)$ threshold one needs *additive* relations.

6.2 Notation for the theorem

All symbols are defined here explicitly.

- $\{P_j\}_{j=1}^d$: the encoding generators (for us, (25)), P_j Hermitian with $\text{spec}(P_j) \subseteq \{\pm 1\}$.
- $\mathcal{H}_{\lambda_0}$: the symmetry sector containing ρ_0 and preserved by the whole circuit (guaranteed by equivariance).
- $F \subseteq \{1, \dots, d\}$: a chosen subset of parameter indices whose generators **mutually commute**, $[P_j, P_{j'}] = 0$ for $j, j' \in F$, and which are applied **contiguously** within each Trotter layer (as the bracketed block in (26)). Commutativity guarantees a joint eigenbasis; contiguity guarantees no other gate intervenes to change the joint eigenvalue pattern mid-block.
- $\epsilon_{s,j} \in \{\pm 1\}$: the eigenvalue branch selected at layer s for generator j on the *ket* side; $\epsilon'_{s,j}$ the same on the *bra* side.
- $u_{s,j} \in \mathbb{F}_2$: the *bit form* of the branch, defined by $\epsilon_{s,j} = (-1)^{u_{s,j}}$. Thus $u = 0$ means eigenvalue $+1$ and $u = 1$ means eigenvalue -1 .
- $c \in \mathbb{Z}^F$: an integer coefficient vector defining an additive relation; $\kappa \in \mathbb{Z}$ the constant it equals.
- $h \in \mathbb{F}_2^F$: the indicator vector of a subset $S = \{j : h_j = 1\}$ defining a multiplicative relation.
- $\mathbb{F}_2$: the field $\{0, 1\}$ with $1 + 1 = 0$; “ $\cdot$ ” between $\mathbb{F}_2$ vectors is the mod-2 inner product.

6.3 Statement

Theorem 6.1 (Symmetry-induced frequency sublattice). *Let $U(x, \alpha)$ be Pauli-encoded with generators $\{P_j\} \subset \rho(G)'$, G abelian, and let ρ_0 and $\mathcal{O}$ be supported in $\mathcal{H}_{\lambda_0}$. Let F be a commuting, contiguously applied family as above. Suppose that, as operators restricted to $\mathcal{H}_{\lambda_0}$,*

$$(i) \text{ additive relations: } \sum_{j \in F} c_j^{(a)} P_j = \kappa_a \mathbb{I}, \quad a = 1, \dots, k, \quad (31)$$

$$(ii) \text{ multiplicative relations: } \prod_{j \in F} P_j^{h_j^{(b)}} = \eta_b \mathbb{I}, \quad \eta_b = \pm 1, \quad b = 1, \dots, p. \quad (32)$$

Then the frequency support satisfies $\Omega \subseteq \Lambda$, with

$$\Lambda = \left\{ l \in 2\mathbb{Z}^d : \underbrace{c^{(a)} \cdot l = 0 \ \forall a}_{k \text{ integer hyperplanes}}, \underbrace{h^{(b)} \cdot \left(\frac{l}{2} \bmod 2\right) = 0 \ \forall b}_{p \text{ parity checks over } \mathbb{F}_2} \right\}. \quad (33)$$

6.4 Proof, line by line

Step 0 (the sector is preserved). $\rho_0, \mathcal{O} \in \mathcal{B}(\mathcal{H}_{\lambda_0})$ and $U \in \rho(G)'$, so $U\mathcal{H}_{\lambda_0} = \mathcal{H}_{\lambda_0}$. Every intermediate state appearing in the expansion of §3.2 therefore lies in $\mathcal{H}_{\lambda_0}$, and only branch assignments realisable *inside the sector* contribute to b_l .

Step 1 (branch assignments are constrained inside the sector). Fix a layer s . Because the generators in F mutually commute and are applied contiguously, the state entering the block can be decomposed in the joint eigenbasis of $\{P_j\}_{j \in F}$ *restricted to $\mathcal{H}_{\lambda_0}$* , and the branch vector $(\epsilon_{s,j})_{j \in F}$ is exactly the joint eigenvalue pattern of one such basis vector. Consequently, any operator identity valid on $\mathcal{H}_{\lambda_0}$ translates directly into a constraint on the admissible patterns.

Step 2 (additive relations $\Rightarrow$ hyperplanes). Apply (31) to a joint eigenvector $|\chi\rangle \in \mathcal{H}_{\lambda_0}$ with pattern $(\epsilon_{s,j})$:

$$\left(\sum_{j \in F} c_j^{(a)} P_j \right) |\chi\rangle = \left(\sum_{j \in F} c_j^{(a)} \epsilon_{s,j} \right) |\chi\rangle \stackrel{(31)}{=} \kappa_a |\chi\rangle \implies \sum_{j \in F} c_j^{(a)} \epsilon_{s,j} = \kappa_a.$$

This holds *at every layer s* , and identically on the bra side with ϵ' . Summing over $s = 1, \dots, L$ and using the definition (10) of the counters:

$$\sum_{j \in F} c_j^{(a)} \Lambda_j = L\kappa_a, \quad \sum_{j \in F} c_j^{(a)} \Lambda'_j = L\kappa_a.$$

Subtracting, and recalling $l = \Lambda - \Lambda'$ from (12):

$$\sum_{j \in F} c_j^{(a)} l_j = L\kappa_a - L\kappa_a = 0.$$

The constant κ_a cancels — which is why the constraint is a hyperplane *through the origin*, independent of the sector label.

Step 3 (multiplicative relations $\Rightarrow$ parity checks). Apply (32) to the same eigenvector, with $S = \{j : h_j^{(b)} = 1\}$:

$$\left(\prod_{j \in S} P_j\right)|\chi\rangle = \left(\prod_{j \in S} \epsilon_{s,j}\right)|\chi\rangle \stackrel{(32)}{=} \eta_b |\chi\rangle \implies \prod_{j \in S} \epsilon_{s,j} = \eta_b.$$

Convert to bits via $\epsilon_{s,j} = (-1)^{u_{s,j}}$:

$$\prod_{j \in S} (-1)^{u_{s,j}} = (-1)^{\sum_{j \in S} u_{s,j}} = \eta_b \implies \sum_{j \in S} u_{s,j} \equiv \nu_b \pmod{2}, \quad \nu_b := \begin{cases} 0, & \eta_b = +1, \\ 1, & \eta_b = -1, \end{cases}$$

and identically $\sum_{j \in S} u'_{s,j} \equiv \nu_b$ on the bra side.

Now relate bits to the frequency increment. Per layer, define $\delta l_{s,j} := \epsilon_{s,j} - \epsilon'_{s,j} \in \{-2, 0, +2\}$. Then

$$\delta l_{s,j} \neq 0 \iff \epsilon_{s,j} \neq \epsilon'_{s,j} \iff u_{s,j} \neq u'_{s,j} \iff u_{s,j} \oplus u'_{s,j} = 1,$$

and since $\frac{\delta l_{s,j}}{2} \in \{-1, 0, +1\}$ with $-1 \equiv 1 \pmod{2}$,

$$\frac{\delta l_{s,j}}{2} \equiv u_{s,j} \oplus u'_{s,j} \pmod{2}. \quad (34)$$

Summing (34) over $j \in S$ and using the two bit constraints:

$$\sum_{j \in S} \frac{\delta l_{s,j}}{2} \equiv \sum_{j \in S} u_{s,j} + \sum_{j \in S} u'_{s,j} \equiv \nu_b + \nu_b \equiv 0 \pmod{2}.$$

Finally sum over layers $s = 1, \dots, L$; since $l_j = \sum_s \delta l_{s,j}$ and reduction mod 2 is additive,

$$\boxed{\sum_{j \in S} \frac{l_j}{2} \equiv 0 \pmod{2}.}$$

Step 4 (evenness). $l \in 2\mathbb{Z}^d$ was established in (13), which is what makes $l_j/2$ an integer and the mod-2 statement well posed. $\blacksquare$

Remark 6.2 (Why the constant cancels in one case but not the other). In Step 2 the sector-dependent constant κ_a cancels between ket and bra, giving a homogeneous constraint. In Step 3 the sign η_b likewise cancels ($\nu_b + \nu_b \equiv 0$). In both cases Λ is a genuine *subgroup* of $\mathbb{Z}^d$, not merely a coset — which is what makes Hermite-Normal-Form reconstruction well defined if one later wants to identify Λ from samples.

7 Instantiation for the $\mathbb{Z}_2$ LGT

7.1 Choosing the commuting family F

Take

$$F = \{Z_v\}_{v \in V} \cup \{X_e\}_{e \in E}, \quad |F| = |V| + |E|. \quad (35)$$

These mutually commute: Z_v and $Z_{v'}$ act on distinct site qubits; X_e and $X_{e'}$ on distinct link qubits; and site/link qubits are disjoint. They are applied contiguously in (26), as required.

The hopping generators h_e are *not* in F : h_e contains Z_e , which anticommutes with X_e . Consequently the $|E|$ hopping coordinates $l_e^{(h)}$ are **unconstrained** by the relations below. This is an honest limitation and we record it explicitly.

7.2 The additive relation: total charge

On the sector $\lambda_0 = (q, \{g_v\})$ we have $Q = \sum_{v \in V} Z_v = q \mathbb{1}_V$. In the language of (31) this is $c = \mathbf{1}_V$ (the all-ones vector on the mass coordinates, zero elsewhere) and $\kappa = q$. Theorem 6.1, Step 2, gives

$$\boxed{\sum_{v \in V} l_v^{(m)} = 0.} \quad (36)$$

One codimension. This is the only additive relation available in the $\mathbb{Z}_2$ model, and the reason is structural: $\mathbb{Z}_2$ Gauss's law is a product, not a sum.

7.3 The multiplicative relations: Gauss's law

On the sector, $G_v = Z_v \prod_{e \ni v} X_e = g_v \mathbb{1}_V$ for every v . In the language of (32), $h^{(v)}$ is the indicator of $\{v\} \cup \delta(v)$ — the mass coordinate of v together with the electric coordinates of all edges incident to v — and $\eta_v = g_v$. Step 3 gives, for each $v \in V$,

$$\boxed{\frac{l_v^{(m)}}{2} + \sum_{e \ni v} \frac{l_e^{(g)}}{2} \equiv 0 \pmod{2}.} \quad (37)$$

Independence. Collect the $|V|$ checks into a matrix over $\mathbb{F}_2$, with columns ordered (mass coordinates | electric coordinates):

$$H_{\text{check}} = [I_{|V|} \mid B] \in \mathbb{F}_2^{|V| \times (|V| + |E|)},$$

where B is the vertex–edge incidence matrix. The left block is the identity, so $\text{rank}_{\mathbb{F}_2} H_{\text{check}} = |V|$: all $|V|$ checks are independent. Their common kernel is a linear code of dimension $(|V| + |E|) - |V| = |E|$.

Reading (37) physically. It says: *the parity of the mass-sector frequency at v is slaved to the parities of the electric-sector frequencies on the edges touching v .* This is Gauss's law transcribed into frequency space: a unit of charge frequency at a vertex must be matched by flux frequency on the incident links.

7.4 Counting: derivation of the reduction factor

We now derive the bound quoted in the summary, factor by factor. Let $\mathcal{L} = \{-2L, \dots, 2L\}^d$ be Barthe's ambient index set, $|\mathcal{L}| = (4L + 1)^d$.

Factor 1 — parity (from (13), no symmetry needed). Only even components can be non-zero. For a coordinate with $r_j = 1$ the admissible count is $2L + 1$ out of $4L + 1$; for $r_j = 2$ it is $4L + 1$ out of the wider ambient $8L + 1$. Taking Barthe's uniform ambient $(4L + 1)^d$ as the reference,

$$\frac{|\{\text{even } l\}|}{|\mathcal{L}|} = \left(\frac{2L + 1}{4L + 1}\right)^d \xrightarrow{L \text{ large}} 2^{-d}. \quad (38)$$

Factor 2 — Gauss parity checks. Each independent $\mathbb{F}_2$ parity check on a set of free bits halves the number of admissible bit patterns. The bits here are $(l_v^{(m)}/2 \bmod 2, l_e^{(g)}/2 \bmod 2)$, and by the rank computation above there are exactly $|V|$ independent checks. Hence

$$\text{factor} = 2^{-|V|}. \quad (39)$$

(This is exact, not asymptotic: the checks are linear and independent, so the solution set is a coset of a subspace of index $2^{|V|}$.)

Factor 3 — the charge hyperplane. We must count the fraction of admissible mass-coordinate vectors satisfying (36). Set $t_v := l_v^{(m)}/2 \in \{-L, \dots, L\}$, so that (36) reads $\sum_v t_v = 0$, with each t_v ranging freely over $2L+1$ values. Drawing the t_v independently and uniformly, each has mean 0 and variance

$$\sigma_1^2 = \frac{1}{2L+1} \sum_{t=-L}^L t^2 = \frac{L(L+1)}{3}.$$

By the local central limit theorem, the sum $S = \sum_v t_v$ is approximately Gaussian with standard deviation $\sigma = \sigma_1 \sqrt{|V|}$, and the probability of the single value $S = 0$ is

$$\Pr[S = 0] \simeq \frac{1}{\sigma \sqrt{2\pi}} = \Theta\left(\frac{1}{L\sqrt{|V|}}\right).$$

Hence

$$\text{factor} = \Theta\left(\frac{1}{L\sqrt{|V|}}\right). \quad (40)$$

Combining. Factors (38) and (39) act on disjoint structures (integer range versus mod-2 parity) and (40) is a further integer constraint; assuming approximate independence between the parity checks and the hyperplane (exact for (38)–(39), heuristic for the interaction with (40)),

$$\boxed{\frac{|\Omega|}{(4L+1)^d} \lesssim \underbrace{2^{-d}}_{\text{parity}} \cdot \underbrace{2^{-|V|}}_{\text{Gauss}} \cdot \underbrace{\Theta\left(\frac{1}{L\sqrt{|V|}}\right)}_{\text{charge}}}. \quad (41)$$

How to read (41). The first two factors are *exponentially large numbers* — for a chain with $|V| = |E| = N_s$ we have $d = 3N_s$, so the combined saving is 2^{-4N_s} . That is a genuine and substantial reduction in the number of Hadamard tests, which dominates Barthe’s training cost $\tilde{\Theta}(m^7/\varepsilon^4)$. But it is a *constant-per-coordinate* saving: the scaling remains $|\Omega| = \Theta(c^d)$ with $c = (4L+1)/2$ rather than $4L+1$. **The exponential in d is not removed, only its base is reduced.** Only the charge constraint (40) genuinely removes a dimension, and there is exactly one of those.

7.5 The pure-gauge limit and the cycle space

Something structurally interesting happens when the matter is frozen (static charges, or the pure $\mathbb{Z}_2$ gauge theory). The mass coordinates disappear, $G_v = \prod_{e \ni v} X_e$, and (37) becomes

$$\sum_{e \ni v} \frac{l_e^{(g)}}{2} \equiv 0 \pmod{2} \quad \forall v \in V. \quad (42)$$

Define $y \in \mathbb{F}_2^E$ by $y_e := \frac{l_e^{(g)}}{2} \bmod 2$. The condition is $By = 0$ with B the $\mathbb{F}_2$ incidence matrix, i.e.

$$\boxed{y \in \ker_{\mathbb{F}_2} B = Z_1(\Gamma; \mathbb{F}_2) = \text{the cycle space of } \Gamma, \quad \dim Z_1 = |E| - |V| + 1 = b_1(\Gamma).} \quad (43)$$

Unpacking the topology, for non-specialists. Over $\mathbb{F}_2$, a set of edges $y \subseteq E$ is called a *1-chain*; its *boundary* ∂y is the set of vertices met an *odd* number of times. A *cycle* is a chain with empty boundary: every vertex is met an even number of times. The cycle space is the vector space of all such y ; its dimension is the *first Betti number* $b_1(\Gamma) = |E| - |V| + 1$ for a connected graph, which counts the **independent loops**. A tree has $b_1 = 0$; a single ring has $b_1 = 1$; adding a chord adds one.

Physical reading. Gauss’s law says electric flux lines can only end on charges. Remove the charges and flux lines must *close into loops*. The frequency parity pattern y is an electric flux configuration mod 2, so it must be a closed loop — a cycle. The admissible frequency parities of the model are literally the loops of the input graph. This is also why the Wilson loop of §5.5 is the natural observable: it is the operator that measures exactly these objects.

Honest caveat. This remains an *index* reduction, not a rank reduction: it constrains $l^{(g)}/2$ only modulo 2, cutting the count by $2^{|V|-1}$ but leaving $\text{rank } \Lambda = |E|$. The appearance of $b_1(\Gamma)$ here is a genuine and encouraging structural signal, but it is not yet the dimension collapse we want. Section 8 explains what would make it one.

8 Assessment and outlook

8.1 What the construction delivers

1. **The equivariance dilemma is resolved.** Because $G = U(1)_Q \times \mathbb{Z}_2^{|V|}$ acts trivially on lattice labels, the classical input x is entirely unconstrained: it may be the edge set of Γ , a bond-disorder pattern entering through fixed-angle gates $\prod_v e^{i\pi x_v Z_v/2}$, or both. Barthe’s concept class survives intact.
2. **Your Fourier theorem’s hypotheses now genuinely hold.** G abelian $\Rightarrow d_\lambda = 1$; sector-supported ρ_0 is automatically in the commutant (§5.4). The Schur decoupling is legitimate, and the spurious d_λ prefactor is gone.
3. **Exact Pauli encoding, at a cost of $\leq 2 \times$ in registers and gates (§5.3),** with the aliasing hazard identified and fixed.
4. **An exponential constant-factor saving** of $2^{-(d+|V|)}$ in the number of Hadamard tests (41), which is the dominant term in the training budget.
5. **Everything is checkable classically in polynomial time.** Because $\mathcal{S} = \langle G_v \rangle$ is a stabiliser group, the questions “is P_j a scalar on the sector?” and “what is the relation lattice?” are $\mathbb{F}_2$ Gaussian-elimination problems of size $O(n^3)$. No exponential Galerkin projection onto a $d_X \times d_X$ subspace is required.

8.2 What it does not deliver, and why

The threshold $d = O(\log n)$ is unchanged. The reason is now precise and is, we believe, the main scientific content of this report:

$$\begin{aligned} \mathbb{Z}_2 \text{ Gauss law is a } \mathbf{product} \text{ of Paulis} &\Rightarrow \text{multiplicative relation} \Rightarrow \mathbf{index} \text{ reduction} \Rightarrow \\ &\text{constant-factor gain.} \\ U(1) \text{ Gauss law is a } \mathbf{sum} \text{ of integer-valued fields} &\Rightarrow \text{additive relation} \Rightarrow \mathbf{rank} \\ &\text{reduction.} \end{aligned}$$

Explicitly, in a $U(1)$ lattice gauge theory the electric field E_e has integer spectrum and Gauss's law reads

$$\sum_{e \ni v} \pm E_e = \rho_v,$$

one *linear* equation per vertex. These are of the form (31), and $|V| - 1$ of them are independent (the sum of all of them is trivial). By Theorem 6.1 Step 2 the rank of the electric-sector frequency lattice collapses:

$$\text{rank } \Lambda_{\text{electric}} = |E| - (|V| - 1) = b_1(\Gamma). \quad (44)$$

A collapse from $|E|$ to $b_1(\Gamma)$ is a genuine *dimension* reduction: it is exactly what would allow a model with $d = \text{poly}(n)$ nominal parameters to behave, for FCE purposes, like a model with $d_{\text{eff}} = b_1(\Gamma)$. On a graph with few independent loops — for instance a tree plus $O(\log n)$ chords — this puts $d_{\text{eff}} = O(\log n)$ and recovers Barthe's efficient regime while keeping $\text{poly}(n)$ physical parameters.

8.3 Recommended role of each model

- **$\mathbb{Z}_2$ model: validation platform.** Finite-dimensional links (one qubit each, no truncation), exactly commuting blocks (Proposition 5.1(iii): no Trotter error inside the hopping block), stabiliser structure (all classical checks polynomial). Use it to verify the register sizing, the parity and Gauss constraints (36)–(37), and the counting (41) numerically against brute-force Fourier extraction on small graphs.
- **$U(1)$ model: separation target.** The additive Gauss law is what produces the rank collapse.

8.4 Open items, stated as risks

- R1. Λ is classically computable, hence cannot itself produce a separation.** Everything in §7 follows from $\mathbb{F}_2$ elimination on $\mathcal{S}$ in polynomial time. A classical learner can restrict its own feature map to the same Λ . Symmetry-induced sparsity makes *both* learners cheaper; it does not open a gap. For the hidden-subgroup framing to become load-bearing, the lattice must depend on something the learner does *not* know — a physical hidden variable such as a flux/holonomy sector, not an architectural symmetry.
- R2. Hardness needs a richer input space.** On a one-dimensional chain with constant evolution time, Lieb–Robinson bounds and matrix-product-state methods simulate the dynamics classically regardless of symmetry. Any claim of separation requires x to range over graphs rich enough to embed a BQP-hard instance (two-dimensional or arbitrary connectivity). We have therefore written everything for a general graph Γ from the outset; the chain is a debugging special case.

- R3. Input-dependent sublattice.** If x is the edge set, then $\mathcal{S} = \mathcal{S}(x)$ and $\Lambda = \Lambda(x)$. This is desirable — it is what makes $b_1(\Gamma(x))$ meaningful — but it means the LASSO design matrix has an x -dependent support, and the regression must be reformulated over the union $\bigcup_t \Lambda(x_t)$. This has not yet been analysed.
- R4. $U(1)$ truncation and the electric term.** The rank- b_1 claim must be checked to survive (a) truncation of the infinite-dimensional link Hilbert space and (b) Trotterisation of the $U(1)$ electric term $\sum_e E_e^2$, which is *quadratic* in the conserved field and therefore not obviously compatible with a ± 1 -spectrum Pauli encoding. We regard this as the first technical obstacle to attack.
- R5. Contiguity hypothesis.** Theorem 6.1 assumes the commuting family F is applied as an uninterrupted block within each layer. This is satisfied by (26) but must be preserved by any compiler or hardware-aware re-ordering.